Building Resilience Index Model

Property Resilience Letter Rating + Flood Resilience Score

MKM Research Labs

Version 1.0 — 15-May-2026

David K Kelly

Legal Notice

PROPRIETARY AND CONFIDENTIAL

Contents

1 Executive Summary

This document presents the mathematical framework, calibration, and validation of the Building Resilience Index (BRI) Model implemented in `src/port/rand/thames/property/property_random/resilience.py`. The model produces two related-but-distinct outputs from a single resilience checklist:

1. **BRI letter rating** (AA / AA+ / A / A+ / B / B+ / NR) derived from a weakest-link aggregation across applicable hazards (Flood, Wind, Fire, Seismic). A `+` modifier is awarded when the operational continuity dimension is also met. The letter scale follows the IFC Building Resilience Index public methodology.
2. **Flood resilience score** (1–100) computed per the *BRI Resilience Function Specification* (April 2026 draft) using a weighted 13-measure model with regime-specific weight multipliers (pluvial / fluvial / coastal), four soft-cap rules, and a linear damage modifier $M(S, r) = 1 - \alpha_r(S - 1)/99$ intended to scale an existing flood damage function.

The model is designed for property-level resilience disclosure, portfolio analytics, and (Phase 2) integration with the existing depth-damage flood loss estimator to produce mitigation-adjusted loss-given-hazard estimates.

Key risk: The resilience checklist 5-level vocabulary (*Not assessed* / *Partial* / *Meets minimum* / *Enhanced* / *Verified*) drives both the letter rating and the score. Inter-rater disagreement on level assignment would propagate directly to both outputs. Validation should include inter-rater agreement studies on a sample portfolio before the model enters production for assessment work.

Calibration status: Thames-specific. Section weights, rating thresholds, and α values reflect the dominant fluvial flood hazard. Non-Thames catchments require a separate calibration. The spec author flagged the underlying α and weight values as "first implementation, not final scientific calibration".

2 Model Purpose and Scope

2.1 Purpose

The BRI Model provides standardised resilience assessment for individual properties. It serves four primary functions:

- **Disclosure:** A human-readable letter grade (AA / A / B / NR) for reporting to lenders, insurers, and policymakers.
- **Portfolio analytics:** A continuous 0–1 score for aggregation, stratification, and tail-risk analysis across property portfolios.
- **Mitigation accounting:** A damage modifier $M(S, r)$ that converts mitigation measures into a multiplicative adjustment on baseline flood damage, supporting loss-given-hazard estimation.
- **Per-hazard transparency:** Independent sub-ratings for Flood, Wind, Fire, and Seismic so consumers can identify the hazard driving an unfavourable overall rating.

2.2 Scope

- **In scope:** Resilience checklist scoring; per-hazard sub-rating; weakest-link overall rating; "+" continuity modifier; flood-spec weighted-measure score with regime multipliers and soft caps;

linear damage modifier.

- **Out of scope:** Hazard intensity estimation (covered by the Storm Intensity and GEV Hazard models); depth-damage curve (covered by the Flood Risk model — Phase 2 integration target); insurance pricing.
- **Upstream dependencies:** `PropertyHeader.PropertyAttributes` (`PropertyPeriod`, `PropertyConditio` `PropertyHeader.Construction` (`FloorLevelMeters`, `BasementPresent`); `PropertyHeader.RiskAssessme` (`EAFloodZone`, `FloodRiskType`); `HistoryAndIncidents.FloodEvents.FloodDamageSeverity`.
- **Downstream consumers:** Governance reporting (MRC dashboards), property report PDFs, future integration with depth-damage flood loss estimator.

2.3 Co-locating Two Methodologies

The module deliberately hosts two related but distinct scoring paradigms.

- The **letter rating** is categorical and weakest-link-aggregated, mirroring IFC BRI public methodology. Its design intent is auditability and comparability across portfolios.
- The **flood score** is continuous and built for damage-modifier use, per the BRI Resilience Function Specification. Its design intent is to retain mitigation-quality detail that a letter rating loses.

Both consume the same resilience checklist input. Co-locating them avoids duplication and keeps the calibration tables in one place.

3 Mathematical Framework

3.1 Compliance Factor

Each resilience checklist field i has a value in one of three forms:

- **5-level enum:** *Not assessed* (0.0), *Partial* (0.4), *Meets minimum* (0.7), *Enhanced* (0.9), *Verified* (1.0) — the dominant form across the checklist.
- **Strategy enum** (`BasementFloodStrategy`): four-valued enum with its own credit map.
- **Continuous** (`FloorLevelMeters`): mapped to a 0–1 compliance via a banded elevation table (Section 3.3).

The compliance factor $c_i \in [0, 1]$ is the credit assigned to field i.

3.2 BRI Letter Rating

3.2.1 Section Score

For each of the five resilience sub-sections $S \in$ {BuildingAssessment, SiteAndDrainage, FloodProtection, FireProtection, ContinuityMeasures} with field weights w_i^S (defaulting to 1.0; critical-control overrides up to 2.0):

$$\text{score}_S = \frac{\sum_{i \in S} w_i^S c_i}{\sum_{i \in S} w_i^S} \tag{1}$$

3.2.2 Overall Weighted Score

A diagnostic overall score is computed but is *not* used to derive the letter; the letter comes from the per-hazard weakest-link path below.

$$\text{score}_{\text{overall}} = \sum_{S} \omega_S \cdot \text{score}_S \tag{2}$$

where section weights ω_S are calibrated for Thames flood-dominant risk (Flood 0.30, Site/Drainage 0.25, Building 0.20, Continuity 0.15, Fire 0.10).

3.2.3 Per-Hazard Score and Rating

For each hazard $h \in \{\text{Flood, Wind, Fire, Seismic}\}$, a relevance map F_h selects the resilience fields that matter for that hazard. The per-hazard score is:

$$\text{score}_h = \frac{\sum_{i \in F_h} w_i c_i}{\sum_{i \in F_h} w_i} \tag{3}$$

and the per-hazard letter is determined by threshold mapping $\tau : [0,1] \to \{\text{AA, A, B, NR}\}$:

$$\tau(s) = \begin{cases} \text{AA} & s \geq 0.87 \\ \text{A} & 0.62 \leq s < 0.87 \\ \text{B} & 0.38 \leq s < 0.62 \\ \text{NR} & s < 0.38 \end{cases}$$

A hazard with `HazardClass = None` is non-applicable and produces *N/A*, which is excluded from aggregation.

3.2.4 Weakest-Link Overall Letter

The overall letter is the weakest applicable letter:

$$\text{rating}_{\text{overall}} = \arg \min_{h \in \text{applicable}} \tau(\text{score}_h) \tag{4}$$

3.2.5 Continuity "+" Modifier

A `+` suffix is appended when the overall is not NR *and* the `ContinuityMeasures` section score meets the threshold:

$$\text{rating}_{\text{overall}} \leftarrow \text{rating}_{\text{overall}} + \text{"+"} \quad \text{iff} \quad \text{score}_{\text{Continuity}} \geq 0.65$$

3.3 Flood Resilience Score

3.3.1 Thirteen-Measure Weighted Sum

The spec defines 13 measures with base weights summing to 100. Each measure $m \in M$ has weight w_m, regime-specific multiplier $\rho_{m,r}$ depending on the flood regime $r \in \{\text{pluvial, fluvial, coastal}\}$, and compliance factor $c_m \in [0,1]$ computed from the CDM record:

$$\tilde{w}_{m,r} = w_m \cdot \rho_{m,r} \tag{5}$$

The raw regime-adjusted score (0–100):

$$S_{\text{raw},r} = 100 \cdot \frac{\sum_m \tilde{w}_{m,r} c_m}{\sum_m \tilde{w}_{m,r}} \tag{6}$$

3.3.2 Banded Elevation Compliance

For `lowest_occupied_floor_elevation` the compliance is read from the elevation margin in metres above the local water reference:

Elevation margin	Compliance
Less than 0 m	0.00
0 to less than 1 m	0.20
1 to less than 3 m	0.50
3 to less than 5 m	0.75
5 to less than 6 m	0.90
6 m or more	1.00

3.3.3 Soft Caps

Four soft-cap rules cap the final score when critical controls fail:

- Lowest occupied floor compliance = 0 ⇒ cap at 40
- Critical systems elevation compliance = 0 ⇒ cap at 60
- Pluvial regime, retainage *and* drainage compliance both = 0 ⇒ cap at 55
- Fluvial/coastal regime, lower-level design *and* water-resistant materials both = 0 ⇒ cap at 65

The final score is:

$$S_{\text{final}} = \max\Big(1, \min\big(S_{\text{raw},r},\ \min_c \text{cap}_c\big)\Big) \tag{7}$$

The lower bound of 1 ensures positive downstream values.

3.3.4 Damage Modifier

The final score maps to a damage modifier scaling baseline damage:

$$M(S, r) = 1 - \alpha_r \cdot \frac{S - 1}{99} \tag{8}$$

with $\alpha = \{0.65 \text{ pluvial}, 0.60 \text{ fluvial}, 0.50 \text{ coastal}\}$. At $S = 1$ no mitigation is applied ($M = 1.0$); at $S = 100$ maximum mitigation is $1 - \alpha_r$.

3.3.5 Adjusted Loss

The intended use (Phase 2 integration) is:

$$\text{LGH}_{\text{adj}} = \text{BaselineDamage}(h, x) \times M(S, r) \tag{9}$$

where h is hazard intensity, x is the existing vulnerability input set, S is the final flood resilience score, and r is the flood regime.

3.3.6 Regime Mapping

The flood regime is derived from `PropertyHeader.RiskAssessment.FloodRiskType`:

FloodRiskType	Regime
Fluvial	fluvial
Pluvial	pluvial
GroundWater	pluvial
Coastal	coastal
Multiple	pluvial *(worst-α default)*

3.4 Score Persistence

For each property the BRI Model writes the following into `ProtectionMeasures.RiskAssessment.GoverningBody`

Field	Value
`BRIRating`	weakest-of applicable hazard letters (with optional "+")
`BRIScore`	mean of the four per-hazard scores
`BRIFloodRating`	per-hazard letter from §3.2
`BRIWindRating`, `BRIFireRating`, `BRISeismicRating`	idem (or N/A)
`BRIFloodScore`	$S_{\text{final}}/100$ from §3.3
`BRIWindScore`, `BRIFireScore`, `BRISeismicScore`	`BRIFloodScore` ± 0.05 (deterministic per PropertyID)
`BRIDate`	date of rating
`BRIRatingVersion`	methodology version (e.g. "BRI v1.6")
`BRIRatingAgent`	verifier (e.g. "Bureau Veritas")

The damage modifier M is returned by `apply_flood_resilience_score` but is *not* persisted in the CDM pending Phase 2 integration with the existing depth-damage model.

4 Input Parameters and Calibration

The model is fully table-driven. All weights, thresholds, and mappings are externalised constants in `src/port/rand/thames/property/property_random/resilience.py` (see Sections 1 and 5 of that file). The Parameter Inventory document lists every parameter with its current value and source line; this section summarises the design rationale.

4.1 Calibration Provenance

- **Section weights** (ω_S): set to reflect Thames fluvial dominance. Flood (0.30) and Site/Drainage (0.25) take precedence; Building Assessment (0.20) and Continuity (0.15) follow; Fire is deprioritised (0.10). Sum to 1.0 (validated by `validate_weights()`).

- **Rating thresholds** (τ): calibrated against the Thames synthetic property generator to hit a 10/20/40/30 target distribution (AA / A / B / NR). Re-tunable via `recalibrate_thresholds_from_scores(` when real loss data becomes available.
- **13-measure spec weights**: taken verbatim from the BRI Resilience Function Specification (Section 3, p. 3). Spec author noted these are "first implementation, not final scientific calibration".
- **Regime multipliers**: taken verbatim from the spec (Section 4, p. 9). Reflect expected per-measure leverage under pluvial vs. fluvial vs. coastal conditions.
- **Soft caps**: taken verbatim from the spec (Section 5, p. 9). Cap values 40, 60, 55, 65 are domain-engineer judgment calls.
- **Damage modifier** α: pluvial 0.65, fluvial 0.60, coastal 0.50, per spec recommended starting values.

4.2 Mapping Spec Measures to CDM Fields

The model reads inputs from existing CDM fields per the binding table below. Nine measures map directly to dedicated fields; four use proxy fields (flagged with †) because the CDM lacks a dedicated equivalent.

Spec Measure (weight)	CDM Source	Proxy?
`lowest_occupied_floor_elevation` (20)	`Construction.FloorLevelMeters` (continuous, elev. table)	
`critical_systems_elevation` (15)	`FloodProtection.ElectricalSystemsAboveFlood` (5-level)	
`site_drainage_topography` (10)	`SiteAndDrainage.OverlandFlowPathsMaintained` (5-level)	
`onsite_retainage_capacity` (10)	`SiteAndDrainage.OnsiteDrainageSizedForDesignStorm` (5-level)	
`permeable_surface_share` (6)	`SiteAndDrainage.PermeableOrRetentionMeasures` (5-level)	
`backflow_protection` (6)	`FloodProtection.BackflowPreventionInstalled` (5-level)	
`lower_level_flood_compatible_design` (8)	`SiteAndDrainage.BasementFloodStrategy` (own enum)	
`water_resistant_materials` (8)	`FloodProtection.PermanentFloodProofingAtEntries` (5-level)	†
`debris_impact_resistance` (5)	`BuildingAssessment.OpeningsWindResistant` (5-level)	†
`roof_drainage_standard` (4)	`BuildingAssessment.RoofRatedForDesignWind` (5-level)	†
`roof_membrane_openings_seal` (3)	`BuildingAssessment.RoofEdgeDetailWindResistant` (5-level)	†
`sump_pumps_backup_power` (3)	`FloodProtection.SumpPump` ∧ `ContinuityMeasures.BackupPowerInstalled` (min)	
`historic_water_area_flag` (2)	`HistoryAndIncidents.FloodEvents.FloodDamageSeverity`	

† *Proxy mappings*: the CDM does not yet capture water-resistant materials, debris impact resistance, roof drainage standard, or roof membrane/openings seal as dedicated fields. We use the closest wind-rated or flood-proofing field as a proxy. See Limitation BRI-L4.

5 Implementation Details

5.1 Module Layout

All resilience logic lives in a single file: `src/port/rand/thames/property/property_random/resilience.py`. The file is organised in six clearly-labelled sections:

1. BRI letter-rating constants (Sections 1 and 2 of the file)
2. BRI letter-rating functions
3. Resilience generator constants (age/condition/zone priors)
4. Resilience generator functions
5. Flood-spec model constants (13 weights, regime multipliers, compliance maps)
6. Flood-spec model functions

Co-locating the constants with the functions that use them keeps the calibration story in one place.

5.2 Helper Module

`src/port/cdm/property/bri.py` provides `apply_bri_rating()` which reads the resilience checklist from a property record and writes the letter rating fields. The flood-spec model is applied by `apply_flood_resilience_score()` in the same `resilience.py` file.

5.3 Builder Integration

`port.src.property.main.builder.BuilderMixin._populate_resilience_and_rating` runs three steps after the schema-walker has built the rest of the property record: (1) `generate_resilience` builds the checklist sub-sections; (2) `apply_bri_rating` stamps the per-hazard + overall letter ratings; (3) `apply_flood_resilience_score` fills the BRI score fields.

5.4 Determinism

Per-property non-flood hazard scores (BRIWindScore, BRIFireScore, BRISeismicScore) are jittered copies of BRIFloodScore ± 0.05. The jitter RNG is keyed by `PropertyID` so that two evaluations of the same property produce identical outputs.

6 Validation and Backtesting

The model is validated through three layers of tests under `tests/port/cdm/`.

6.1 Test Suite Layout

- `test_bri_helper.py` (8 tests) — `apply_bri_rating` writes the correct fields, leaves score fields blank where appropriate, handles missing `HazardProfile`.
- `test_bri_aggregation.py` (34 tests) — letter-rating math: `_field_credit` for booleans and enums, `score_section`, `score_hazard`, weakest-link logic, "+" modifier, calibration helpers.
- `test_resilience_generator.py` (21 tests) — generator behaviour: level mapping, age-bias correctness, condition multiplier, flood-zone uplift, BasementFloodStrategy invariants, distribution sanity check.
- `test_consistency_fixes.py` (37 tests) — builder post-step consistency rules.
- `test_schema_invariants.py` (30 tests) — structural invariants on `PROPERTY_SCHEMA`.
- `test_flood_resilience_model.py` (68 tests) — new flood-spec model: constants, elevation banding, compliance for all 13 measures, soft caps, regime mapping, damage modifier α boundaries, write-back semantics.

6.2 Sanity Checks

Beyond unit tests, the module exposes `sanity_check_distribution()` which generates a synthetic Thames property population and reports the actual vs. target rating distribution. The Thames target is 10/20/40/30 (AA/A/B/NR).

6.3 Automated Test Results

The Building Resilience Index Model is covered by 198 tests across six files under `tests/port/cdm/`. All currently pass. Detailed pass/fail counts and execution time are populated by the test-results generator.

Test File	Cases
`test_bri_helper.py`	8
`test_bri_aggregation.py`	34
`test_resilience_generator.py`	21
`test_consistency_fixes.py`	37
`test_schema_invariants.py`	30
`test_flood_resilience_model.py`	68
Total	**198**

7 Sensitivity Analysis

Table 1: Mean flood resilience score by construction period

Period	Mean score (1-100)
Pre-1919	58.6
1919-1944	54.8
1945-1975	45.2
1976-1999	63.1
2000-2008	75.0
2009-Present	83.1

Table 2: Mean flood resilience score by EA flood zone

Zone	Mean score (1-100)
Zone 1	58.4
Zone 2	63.1
Zone 3a	69.9
Zone 3b	74.4

Table 3: Mean flood resilience score by property condition

Condition	Mean score (1-100)
Excellent	74.4
Good	66.8
Fair	63.1
Poor	51.8
Very poor	38.3

Table 4: Mean flood resilience score by flood regime (FloodRiskType)

Regime	Mean score (1-100)
Pluvial	62.2
Fluvial	63.1
Coastal	63.1

8 Model Limitations and Known Weaknesses

The model is structurally faithful to its design intent but is calibrated for *indicative* rather than *actuarially accurate* use. Several limitations are inherited from the spec author's own caveats.

8.1 Known Limitations

BRI-L1: Distribution shape. The current flood resilience score distribution is bell-centred (mean ≈ 58, median ≈ 57) rather than the design-target right-skewed shape with a heavy left tail of low-quality stock. Generator priors will need re-tuning.

BRI-L2: No location/hazard context normalisation. A Zone 1 property and a Zone 3b property with identical resilience measures currently receive identical scores. The IFC BRI methodology supports context-normalisation (high-hazard sites need more measures for the same rating), but it is not yet implemented. *Design under review.*

BRI-L3: Damage modifier not integrated. $M(S, r)$ is computed and returned by `apply_flood_resilience_` but is not yet wired into the existing depth-damage flood loss estimator. Phase 2.

BRI-L4: Four proxy mappings (BRI-A7). `water_resistant_materials`, `debris_impact_resistance`, `roof_drainage_standard`, and `roof_membrane_openings_seal` use proxy CDM fields. Aggregate weight at risk: 20 of 100 spec points.

BRI-L5: Thames-specific calibration. Weights and thresholds do not generalise to other catchments without re-tuning. Per-catchment copies of `resilience.py` required when other catchments are onboarded.

BRI-L6: Confidence score not implemented. The spec defines a separate confidence indicator combining verification, evidence quality, recency, and field completeness. Not yet implemented.

8.2 Known Failure Modes

- All-Verified resilience checklist produces score near 100 — unrealistic if no real-world property scores that high. Soft caps mitigate but do not eliminate.
- Properties with `HazardClass = None` across all hazards produce overall rating from the fallback weighted-score path, which differs subtly from the weakest-link path used when at least one hazard is applicable.
- `FloorLevelMeters` at or below 0 triggers the `lowest_occupied_floor_elevation` soft cap at 40 — correct behaviour but coarse.
- Regime multipliers and soft caps interact non-trivially: a pluvial-regime property with both retainage and drainage at compliance = 0 is capped at 55 regardless of other strengths.
- PropertyID-keyed jitter RNG for non-flood hazard scores means two properties with the same

flood score receive different sibling scores, which is intentional but can surprise consumers expecting hash-stability.

9 Change History

Version	Date	Description
1.0	2026-05-15	Initial implementation. BRI letter rating (weakest-link across hazards with "+" continu